

6	(a) (i)	1 total resistance = $0.16 \, \Omega$	A1	
		2 e.m.f. = <i>either</i> $(14 - E)$ <i>or</i> $(E - 14)$	A1	[2]
	(ii)	<i>either</i> $14 - E = 42 \times 0.16$ <i>or</i> $(E - 14) = -42 \times 0.16$ $E = 7.3 \, \text{V}$	C1	
			A1	[2]
	(b) (i)	charge = It $= 12.5 \times 4 \times 60 \times 60$ $= 1.8 \times 10^5 \, \text{C}$	C1	
			A1	[2]
	(ii)	<i>either</i> energy = EQ <i>or</i> energy = Eit <i>either</i> energy = $14 \times 1.8 \times 10^5$ <i>or</i> energy = $14 \times 12.5 \times 4 \times 3600$ $= 2.52 \times 10^6 \, \text{J}$	C1	
			A1	[2]
	(iii)	energy = I^2Rt <i>or</i> Vit <u>and</u> $V = IR$ $= 12.5^2 \times 0.16 \times 4 \times 3600$ $= 3.6 \times 10^5 \, \text{J}$	C1	
			A1	[2]
	(c)	efficiency = $(2.52 \times 10^6 - 3.6 \times 10^5)/(2.52 \times 10^6)$ $= 86\%$	C1	
			A1	[2]